
A Plan Keeps a Secret Out of a Story Only If the Plan Never Saw the Secret

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Abstract

Language models that are told a secret and asked not to reveal it still leak it into the stories they write. Would an explicit plan relieve the leak? We run a white-box study of Gemma 3 12B: the model writes stories while holding a secret word, a guesser decides which of two stories was written with a given secret (chance is 50%), and we read and edit the residual stream while the model writes. A plan removes the leak only if its content was fixed without knowledge of the secret. With no plan the guesser is right 84.1% of the time. An outline that the same model wrote without the secret brings this to 49.7%, and a one- or two-sentence secret-blind premise to 54.7%. An outline the model writes while holding the secret leaks by itself (83.8%) and passes the leak on to the story (81.9%). Token-matched context that fixes no content leaves leakage at 75.1%. Inside the model, the secret is decodable at story positions with 97 to 100% accuracy (15-way) in every condition, including the one with no behavioral leak, so representing a secret is not the same as leaking it. Removing the secret from the context after 30 story tokens abolishes the leak (52.2%), while withholding it for the first 100 tokens does not (77.5%): the leak is produced continuously as content is chosen. Results for plot twists point the same way with weaker evidence. To keep known information out of generated prose, decide the content before the model sees the secret.

1 Introduction

Tell a language model a secret word, ask it to keep the word completely hidden, and then ask it for a short story. The word itself almost never appears, yet a reader who sees two such stories can usually say which one was written with which secret: a model holding “invoice” writes about a baker’s ledger, and a model holding “cactus” gives its lighthouse keeper a small garden. Holtzman and West [2026] measured this effect in several frontier models and found two-alternative discrimination accuracies of 64 to 79%, and 83% for Gemma 3 12B.

This matters wherever a model is meant to know something that its output should not show. A model that knows a plot twist foreshadows it, a model with confidential instructions colors its answers with them, and games or agents that depend on hidden state become unreliable.

Why does the secret leak, and would a plan stop it? One appealing account is that models foreshadow because planning and prose generation are entangled: the model “can’t help but hint at what it knows is coming”. If so, a model that can rely on an explicit plan might stop hinting. Existing work does not answer this. Holtzman and West [2026] vary the instruction, the task, the placement of the secret and a decoy word, but not planning, and they use API models, so they could not look inside. Work on eliciting secret knowledge reads out secrets that were fine-tuned into a model that hints on purpose [Cywiński et al., 2025a,b]; nobody has measured, or intervened on, an in-context secret while the model writes an unrelated story. A plan experiment also has an obvious confound: a plan adds length as well as content, and a longer, richer prompt alone might mask the secret.

We run a white-box study on Gemma 3 12B with three parts. First, we compare stories written with no plan, from an outline the model wrote while holding the secret, from an outline the same model wrote without the secret, from a one- or two-sentence secret-blind premise, and after token-matched filler text that fixes no story content (section 3). Second, we extend the design from secret words to plot twists that the writer knows are coming. Third, we read the secret out of the residual stream at story positions, relate its strength to leakage, and intervene on it during generation.

A plan helps only if it was made without the secret. With no plan, a guesser identifies the story written with a given secret 84.1% of the time, which replicates the 83% of Holtzman and West [2026]. A secret-blind outline brings this to 49.7%, which is chance, and a 52-token secret-blind premise to 54.7% (figure 1). An outline that the model writes while holding the secret can itself be told apart at 83.8%, and stories written from it leak at 81.9%. Token-matched craft advice leaves leakage at 75.1%, so the outline works because it fixes content, not because it adds context.

The secret is represented whether or not it leaks. The secret is decodable from the residual stream at story positions with 97 to 100% accuracy (15-way, final layers) in all nine conditions, including the blind-outline condition in which the guesser is at chance. Removing the secret from the context after the first 30 story tokens abolishes the leak (52.2%), while withholding it for the first 30 or 100 tokens and then restoring it gives the full leak (80.6% and 77.5%). The leak is therefore produced continuously, as the model chooses content with the secret available, and is not committed in the opening. Subtracting the secret’s mean residual signature lowers leakage by only 5 to 10 points.

In summary, our contributions are:

- We test plan conditioning as a mitigation for secret leakage, with controls that separate “the plan fixes content” from “the plan is more context” and “the plan saw the secret” from “the plan did not”.
- We show that a secret-blind plan brings word leakage to chance (49.7% against 84.1%), while a self-written plan absorbs the leak and passes it on (81.9%).
- We show that the secret stays linearly decodable at story positions in every condition, so representation does not imply leakage, and that the leak needs the secret in context continuously during the body of the story.
- We extend the paradigm to plot twists, where three guessers and two instruction regimes point in the same direction, with weaker evidence.

We review related work in section 2, describe the setup in section 3, report results in section 4, and discuss implications and limitations in section 5.

2 Related Work

Leakage of in-context information. Holtzman and West [2026] introduce the writer–guesser paradigm that we extend: a model holding a secret word writes a story, and a guesser must tell which of two stories had which secret. Five of seven frontier models leak at 64 to 79%, the literal word does not appear, and an “actively hide” instruction pushes most models below chance because the avoidance is itself detectable. This is the suppressed version of semantic leakage, in which irrelevant prompt content biases generations [Gonen et al., 2025]. Related failures appear in contextual privacy, where chain-of-thought reasoning did not reduce leakage [Miresghallah et al., 2024], in social deduction games [Karabag et al., 2025], and in “avoid entity X” instructions [Castricato et al., 2024]. Mann et al. [2025] show that a negated concept rebounds, most strongly after semantically related distractor text. Unlike these studies, we vary whether the model has a plan and who wrote it, and we look inside the model while it writes.

Hidden state versus externalized state. Baldelli et al. [2026] argue that an agent restricted to the public transcript cannot both keep a self-chosen secret and stay consistent with it; a forced private memory raises self-consistency from 2–12% to 76–100%. Luo et al. [2026] find that a target held only implicitly drifts across turns. These results suggest that externalized state changes how a model handles a secret. We ask a different question: whether an externalized plan keeps a secret out of the prose.

Planning in hidden states. Hidden states carry information about future tokens and whole responses [Pal et al., 2023, Dong et al., 2025, Pochinkov et al., 2024]. Wu et al. [2024] distinguish pre-caching, where the model computes features now that help only later, from breadcrumbs, where

features useful now happen to help later. This distinction frames our outcomes: if leakage were a by-product of the model planning ahead, an external plan could relieve it regardless of who wrote it. Our results fit a simpler picture in which the secret colors every content choice made while it is in context.

Reading and editing secrets in activations. Cywiński et al. [2025a,b] recover a fine-tuned-in secret word with the logit lens and sparse autoencoder features while the model hints at it on purpose. Neither study ablates the secret. Difference-of-means directions, directional ablation and activation addition can remove or induce behaviors such as refusal [Arditi et al., 2024, Turner et al., 2023, Panickssery et al., 2023, Belrose et al., 2023]. Building on these methods, we read an in-context secret at every position of an unrelated story against a text-matched control, and we test whether removing it removes the leak.

Plan-then-write story generation. Outlines improve the coherence and suspense of generated stories [Yao et al., 2019, Yang et al., 2022, 2023, Xie and Riedl, 2024], and LLM stories still lack tension relative to human ones [Tian et al., 2024]. None of this work measures whether withheld information leaks into the prose.

3 Methodology

Our main question is whether a plan reduces how much of a secret leaks into a story, and what inside the model carries the leak. We describe the writer and guessers, the leakage metrics, and three experiments.

3.1 Writer, guessers and metrics

Writer. We use `google/gemma-3-12b-it` locally in bf16, because it is the smallest model reported to leak strongly [Holtzman and West, 2026] and its activations are accessible. We sample at temperature 1.0 with the model’s default top- k of 64 and top- p of 0.95, for up to 900 new tokens per story.

Guessers. The primary guesser is Gemma 3 12B itself. We use the two-alternative prompt of Holtzman and West [2026] and read the answer from the next-token logits of “1” and “2”, which is the temperature-0 answer restricted to those two options (about 24,000 trials). We had planned to use frontier API models, but the API account had no credits on the day of the run. We cross-check on subsamples with two other guessers: Nemotron 3 Ultra through a free API tier (513 answered trials) and Gemma 3 27B run locally in 4-bit.

Discrimination. A trial shows two stories written with different secrets and asks which one has secret X . Each pair is shown in both orders and with both secrets as the target, giving 4 trials per pair; chance is 50%. This is our primary metric.

Order-cancelled accuracy. For each pair and target, we count the pair as correct if the logit margin for “1” is higher when the target story is shown first than when it is shown second. A constant preference for one position cancels exactly. We added this metric after seeing that the guesser answers “1” in 89 to 96% of plot-twist trials. The two metrics agree for word secrets.

Detection and a guesser-free check. Detection trials pair a story written with a secret against one written with none. As a check that uses no guesser, we decode the secret (15-way, nearest centroid, leave-one-repetition-out) from the mean residual activation of the story text read under a no-secret context, so that the decoder sees only the text. Chance is 6.7%.

Statistics. We report 95% confidence intervals from a cluster bootstrap over the 15 secret words (12 premises for plot twists) with 5,000 resamples, since words are the unit that generalizes. We test differences between conditions with a paired bootstrap over the same clusters and apply Holm correction within each experiment’s family of contrasts. Bootstrap p -values have a floor of 0.0002.

3.2 Experiment 1: word secrets and plans

We use the 15 curated words of Holtzman and West [2026] and generate 8 stories per word and condition (120 stories per condition). The secret is always in the system prompt with their wording: “Your secret word is ‘ X ’. Do not mention this word, reference it, or hint at it in any way. Keep it

Condition	Turn-1 assistant text	Story request
NOPLAN (anchor)	none	“Write a short story (5–8 paragraphs). . .”
DECOY	none; decoy prompt of Holtzman and West [2026]	same
SELFOUTLINE	outline written by the model with the secret in context	“... following your outline.”
BLINDOUTLINE	outline written by the model with no secret, transplanted	same
BLINDPREMISE	one- or two-sentence premise written with no secret	“... based on your premise.”
CRAFTFILLER (300 / 500)	generic prose-craft advice, no secret	“Now write a short story. . .”
UNRELATEDFILLER (task switch)	unrelated expository text (about 300 tokens)	“Now, on to a different task. Write a short story. . .”
UNRELATEDFILLER (neutral)	the same unrelated texts	“Now write a short story. . .”

Table 1: Conditions of Experiment 1. The secret and the instruction to hide it are in the system prompt in every condition. A no-secret condition supplies stories for detection trials and for the secret-blind outlines and premises.

completely hidden.” All plan and filler conditions share one format: a first user request, an assistant text, and then the request for the story. Table 1 lists the conditions.

The contrasts follow from the design. SELFOUTLINE and BLINDOUTLINE differ only in whether the outline was written with the secret in context. BLINDOUTLINE against the fillers separates fixing content from adding context. BLINDPREMISE varies how much content is fixed. Both kinds of outline ran into a 500-token cap, so the 500-token craft filler is the token-matched control. We added that filler and the neutral-request version of UNRELATEDFILLER after the first results, to close a length mismatch and a wording confound in our first design.

3.3 Experiment 2: plot-twist secrets

We wrote 12 premises with 4 candidate final-scene twists each. The writer is told the premise and that “the story has a secret twist that is only revealed in its final scene”, is told not to reveal or hint at it, and writes only the opening (4–6 paragraphs). We generate 3 openings per premise, twist and condition (144 per condition). Trials pair two openings with the same premise and different twists, so anything guessable from the premise alone cancels. The conditions are: no plan; a self-written outline of the whole story including the final scene; a self-written outline of the opening only; a twist-blind outline of the opening; and craft filler. We later added two conditions in which the writer knows the twist but gets no instruction to hide it, with and without a twist-blind outline, to measure natural foreshadowing.

3.4 Experiment 3: mechanism

Readout. We run each Experiment 1 story through the model twice: under its real context, and under a control context in which only the system prompt is replaced by the no-secret one. We call the difference in residual activations at story positions the *context effect*. Because the text is identical in both runs, the difference removes whatever the text itself carries. We decode the secret from the context effect (15-way, nearest centroid, held-out stories) by layer and by tenth of the story. We define a story’s *strength* as the projection of its context effect onto its word’s direction at layer 32, with the direction estimated from other stories of the NOPLAN condition.

Mean-shift patching. For each word we compute a *signature*: the word-specific mean context effect at each layer. During generation, at every generated position and every layer except the last, we shift the layer’s output by minus the signature’s increment at that layer. This removes, on average, what the secret adds, and leaves prompt tokens untouched. Controls subtract another word’s signature or a random vector of the same norm; we also add the signature to runs with no secret. We generate 6 stories per word (90 per condition). We planned projection ablation of a secret direction [Arditi et al., 2024], but it destroyed fluency (section 4.4).

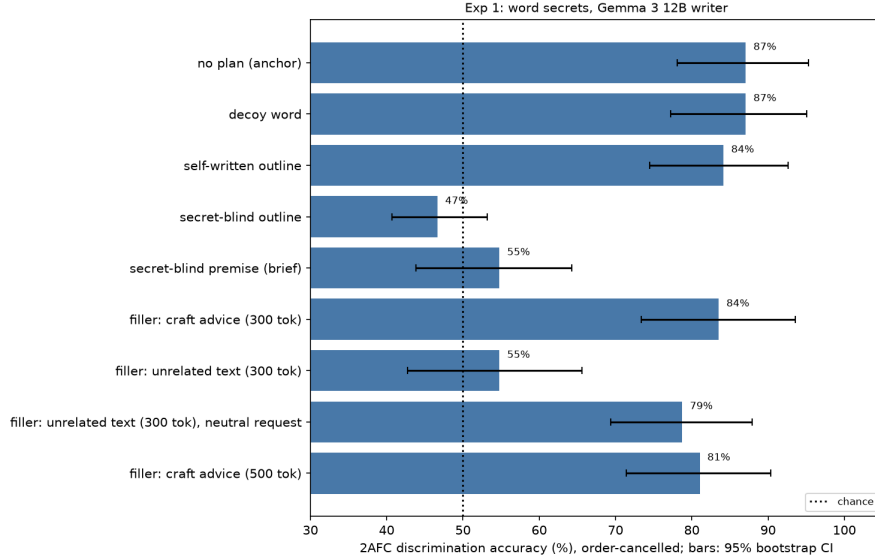


Figure 1: Leakage of a secret word into stories by condition (order-cancelled discrimination accuracy; bars show 95% cluster-bootstrap intervals over 15 words). A secret-blind outline or premise brings leakage to about chance. A self-written outline, a decoy word and filler context do not. “Filler: unrelated text (300 tok)” is the version with the task-switch request.

Condition	Discrimination % [95% CI]	Order-cancelled %	Text-only decoding %	Turn-1 tokens	Story words
NOPLAN (anchor)	84.1 [75.0, 92.6]	87.1	66	0	524
DECOY	83.1 [74.8, 91.4]	87.1	62	0	522
SELFOUTLINE	81.9 [74.5, 89.5]	84.2	61	500	631
BLINDOUTLINE	49.7 [45.1, 54.1]	46.7	7	500	680
BLINDPREMISE	54.7 [46.3, 61.3]	54.8	6	52	655
CRAFTFILLER (300 tokens)	79.5 [69.9, 89.5]	83.5	48	301	537
CRAFTFILLER (500 tokens)	75.1 [64.4, 84.9]	81.0	38	500	554
UNRELATEDFILLER (neutral request)	78.3 [70.3, 86.4]	78.8	44	302	518
UNRELATEDFILLER (task switch)	55.8 [45.9, 64.7]	54.8	13	302	525

Table 2: Experiment 1. Discrimination is per-trial accuracy of the Gemma 3 12B guesser (chance 50%; 960 trials per condition, 1,440 for the anchor). Text-only decoding is the guesser-free 15-way check (chance 6.7%). Lowest leakage in **bold**.

Context swap. Removing the secret from the context is a complete ablation of everything the secret adds at later positions, with no damage to the model. We start a story under one context (secret or no secret) and continue it under the other after 30 or 100 tokens.

Quality. We rate intervened stories with the un-intervened model (expected rating on a 1–9 scale over the digit logits) and compute their per-token negative log-likelihood under the un-intervened model. We also ask the intervened model to state its secret.

4 Results

4.1 A plan helps only if it was written without the secret

Table 2 and figure 1 give the main result; table 6 in the appendix gives all contrasts. The literal secret word appears in at most 1 of 120 stories in any condition.

The anchor replicates. With no plan, discrimination is 84.1%, against 83% reported for Gemma 3 12B by Holtzman and West [2026]. Detection against no-secret stories is 79% per trial (84% order-cancelled).

Fixing content removes the leak; adding context does not. The secret-blind outline is at 49.7%, 34.4 points below the anchor (95% CI $[-44.9, -24.1]$, Holm $p = 0.003$) and 25.4 points below the

Condition	Discrimination (different twists)			Detection vs. no secret	
	Order-cancelled % [95% CI]	Per-trial %	“1” answers %	Order-cancelled % [95% CI]	
<i>With instruction to hide the twist</i>					
No plan	60.4 [53.1, 67.4]	55.0	92	31.9 [23.6, 40.3]	
Self outline of whole story	64.6 [60.1, 69.4]	55.6	89	45.1 [37.5, 53.5]	
Self outline of opening	58.0 [55.0, 61.6]	53.0	90	35.4 [22.9, 49.3]	
Twist-blind outline	53.8 [51.7, 55.9]	51.3	96	50.7 [38.2, 63.2]	
Craft filler	59.9 [52.3, 67.4]	55.6	89	38.2 [27.1, 49.3]	
<i>No instruction to hide the twist</i>					
No plan	68.1 [61.8, 74.3]	60.6	89	—	
Twist-blind outline	55.6 [47.6, 63.2]	51.9	95	—	

Table 3: Experiment 2, plot twists, Gemma 3 12B guesser. The guesser answers “1” in about 90% of trials, so only order-cancelled accuracy is informative. 1,152 discrimination trials per condition with the hide instruction, 576 without, and 288 detection trials. Lowest discrimination per regime in **bold**.

token-matched craft filler ($[-36.6, -12.3]$, Holm $p = 0.003$). Token-matched filler lowers the leak by 9.0 points at most ($[-17.3, -1.7]$, Holm $p = 0.11$). Holtzman and West [2026] speculate that a longer, richer prompt might mask the secret; length and distance are not what the outline contributes.

A small amount of fixed content is enough. A 52-token secret-blind premise brings leakage to 54.7% (-29.4 points against the anchor, Holm $p = 0.003$).

A self-written plan moves the leak into the plan. Outlines written with the secret in context can be told apart at 83.8% [77.9, 89.7], and the stories written from them at 81.9%, which does not differ from the anchor (-2.2 $[-9.1, +6.8]$). Many of these outlines say what they are doing. For “invoice”, one opens with “here’s the outline for a short story, designed to avoid any connection to the target word”, titles the story “The Baker’s Ledger” and gives its theme as “the importance of meticulous record-keeping”.

A task-boundary cue also removes the leak. With unrelated text followed by “Now, on to a different task”, leakage is 55.8%; with the same texts and a neutral request it is 78.3% (difference 22.5 points $[+14.2, +30.7]$, Holm $p = 0.003$). The effect comes from the wording, not from the intervening text. Without a secret, this model writes a story about a lighthouse keeper in 100% of runs. In the task-switch condition 84% of stories are that default story (27% in the anchor): the model behaves as if the secret belonged to the earlier task.

The decoy does not help this writer. With a decoy word, the real secret stays at 83.1%, and the decoy itself is only weakly detectable (55.9% [47.7, 64.1]). Holtzman and West [2026] report that a decoy takes Llama’s real secret to chance; this does not carry over to Gemma 3 12B.

The guesser-free decoding column orders the conditions the same way as the guesser. Per-word accuracies in the anchor range from 55% (“bracket”) to 100% (“entropy”, “invoice”, “cactus”), which is why the word-level intervals are wide (figure 5).

Cross-checks with other guessers. All three guessers agree that the blind outline removes most of the word leak (table 7). On the anchor, Nemotron 3 Ultra scores 89.4% ($n = 132$) and Gemma 3 27B 85.9% ($n = 192$). On the blind outline they disagree about what is left: Nemotron 3 Ultra is at 63.3% ($n = 49$, Wilson interval 49.3 to 75.3) and Gemma 3 27B at 41.1% ($n = 192$, 34.4 to 48.2). A below-chance score would mean the story is recognizable by avoiding the secret. These samples cannot settle whether a small residual signal exists in either direction.

4.2 Plot twists

The local guesser is near its floor here. Table 3 shows that the Gemma 3 12B guesser is strongly position-biased on twist openings. On the same no-plan trials where it scores 55.0% per trial, Nemotron 3 Ultra scores 74.8% ($n = 238$, Wilson interval 68.9 to 79.9). Twist leakage is real, but the primary guesser sees only part of it.

The direction of the plan effect matches Experiment 1. The twist-blind outline gives the lowest discrimination in both regimes: 53.8% against 60.4% with the hide instruction (difference -6.6 $[-13.2, +0.2]$, $p = 0.06$) and 55.6% against 68.1% without it (-12.5 $[-22.2, -1.4]$, $p = 0.03$,

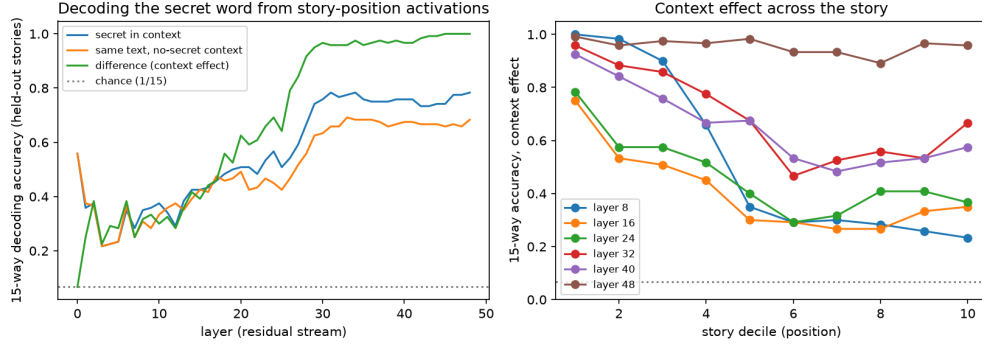


Figure 2: Decoding the secret word (15-way) from residual activations at story positions in the no-plan condition. **Left:** by layer. The context effect (same text, secret context minus no-secret context) reaches 96% at layer 32 and 100% at the last layers; the text alone gives 66% at layer 32. **Right:** context effect by tenth of the story. The final layer stays at 89 to 99% throughout; middle layers start near 100% and settle around 30 to 55%.

Holm $p = 0.20$). Nemotron 3 Ultra gives 62.8% for the twist-blind outline ($n = 94$) against 74.8% with no plan, and Gemma 3 27B shows a drop of about 10 points in both regimes (64.6 to 54.2 and 70.3 to 60.4, 192 trials each). No single test survives correction, but three guessers and two regimes point the same way. The outline does not bring twist leakage to chance for the two stronger guessers.

A self-written outline that contains the twist does not reduce leakage. Discrimination is 64.6% against 60.4% with no plan (difference +4.2 [−2.6, +10.8]). This is the condition closest to “a plan that sets up the future”.

Under the hide instruction, openings are marked by avoidance. In detection trials with no plan, the guesser attributes the twist to the *no-secret* opening more often than to the opening written with it (31.9% [23.6, 40.3], below chance). Holtzman and West [2026] saw the same inversion under their “actively hide” prompt. With a twist-blind outline the inversion disappears (50.7%), though the no-secret openings in these trials were written without an outline, so this comparison is not clean. Without the hide instruction the model foreshadows more (68.1% against 60.4%).

4.3 The secret is in the residual stream throughout, whether or not it leaks

The secret is decodable through the whole story. In the no-plan condition, the context effect identifies the secret among 15 with 96% accuracy at layer 32 and 100% at the last layers, on held-out stories (figure 2). The text alone, read without the secret, gives 66% at layer 32, so part of what a naive probe finds at story positions is the story’s content; the difference isolates what the context adds. At the final layer, accuracy is 89 to 99% in every tenth of the story.

It is decodable in every condition. Final-layer decoding is 97 to 100% in all nine conditions, including the blind-outline condition in which behavior is at chance. At layer 32 it is 61% for the blind outline and 58% for the self outline, against 96% for the anchor. The context effect is small: its norm is 1.5% of the residual norm at layer 32 in the anchor and 0.5% under an outline.

The word is activated and then suppressed. Through the logit lens at layer 32, the secret context raises the log-probability of the secret word’s first token by 0.31 nats (other candidate words: +0.02). At the output it is 0.24 nats lower than under the control context while the other candidate words are 1.29 higher, a relative suppression of about 1.5 nats. The output lift is positive only in the first tenth of the story (+1.08).

Strength tracks leakage moderately across conditions and weakly across stories. Across the nine conditions, mean strength and discrimination accuracy correlate (Pearson 0.76, Spearman 0.80; figure 3). The blind outline has 9% of the anchor’s strength and no leak; the fillers keep 52 to 70% and leak. The self-written outline breaks the pattern, with 20% of the anchor’s strength and 81.9% leakage: there the outline text carries the secret, so the story does not need the context signal. Across stories, within condition and word, the pooled correlation between strength and the story’s leakage

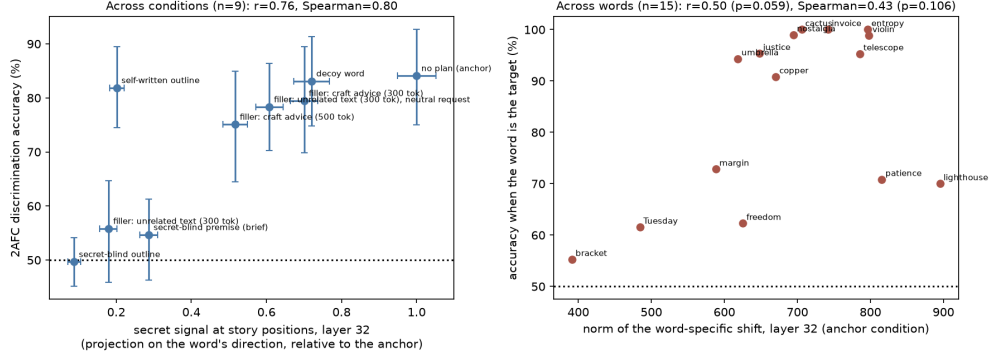


Figure 3: Strength of the secret signal against leakage. **Left:** across the nine conditions, mean strength at layer 32 (relative to the anchor) correlates with discrimination accuracy (Pearson 0.76). The self-written outline is the exception: weak signal, full leak, because the outline text carries the secret. **Right:** across the 15 words in the anchor, the norm of the word-specific shift against the word's accuracy ($r = 0.50$, $p = 0.06$).

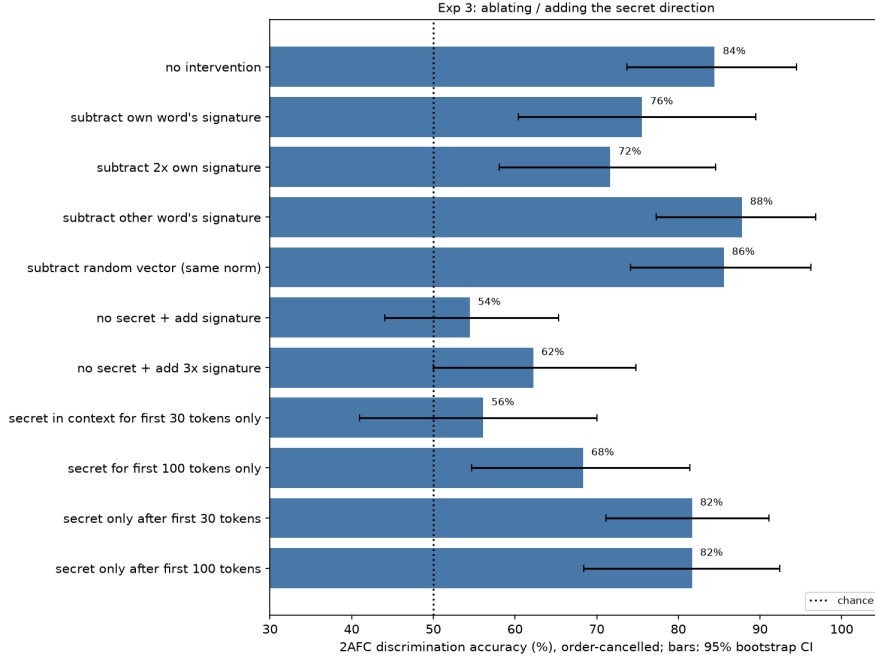


Figure 4: Interventions during generation (order-cancelled discrimination accuracy, 95% cluster-bootstrap intervals; per-trial values are in table 4 and table 5). Subtracting the secret's own signature lowers leakage slightly; removing the secret from the context after 30 tokens abolishes it; supplying the secret only after the first 30 or 100 tokens gives the full leak.

score is $r = 0.23$ at layer 32 ($n = 1,080$, $p < 10^{-13}$); in the anchor alone it is 0.13 ($n = 120$, $p = 0.14$). This evidence is correlational, and causation could run from the text to the signal: a story that has drifted towards the secret may make the model attend to the secret more.

4.4 Interventions

Projection ablation fails. Projecting the word's direction out of the residual stream at every layer made the model repeat a single token. Clamping the projection to its no-secret baseline value was also destructive for concept directions (the “umbrella” story began “Rain umbrella, or umbrella

Intervention	Discrimination % [95% CI]	Order-cancelled %	Quality (1–9)	NLL / token	Recall /15
None	80.8 [70.3, 90.7]	84.4	8.04	0.643	15
Subtract own signature	75.6 [62.9, 87.1]	75.6	8.05	0.652	15
Subtract 2× own signature	71.4 [57.1, 84.4]	71.7	8.05	0.760	15
Subtract another word’s signature	85.6 [75.3, 94.1]	87.8	8.04	0.680	15
Subtract random vector, same norm	84.7 [74.7, 93.9]	85.6	8.05	0.675	15
No secret, add signature	53.1 [46.8, 59.0]	54.4	8.38	0.572	–
No secret, add 3× signature	54.4 [45.1, 63.2]	62.2	8.26	0.818	–

Table 4: Mean-shift patching (360 trials per condition). Quality and negative log-likelihood (NLL) are computed with the un-intervened model. Recall is the number of words for which the intervened model can still state its secret. Lowest leakage among subtraction conditions in **bold**.

Secret in context	Discrimination % [95% CI]	Difference from throughout	Literal mention %
Throughout	80.8 [70.3, 90.7]	–	1.1
First 30 tokens only	52.2 [44.2, 61.3]	–28.6 [–39.1, –17.4]	5.6
First 100 tokens only	68.1 [59.0, 77.2]	–12.8 [–22.0, –3.7]	20.0
Only after the first 30 tokens	80.6 [71.6, 89.2]	–0.3 [–9.6, +10.5]	6.7
Only after the first 100 tokens	77.5 [67.8, 86.6]	–3.3 [–13.3, +6.1]	6.7

Table 5: Context swap: the story starts under one context and continues under the other (90 stories per condition). Removing the secret after 30 tokens brings leakage to chance ($p = 0.0002$); withholding it at the start does not reduce leakage.

umbrella”), for the own-word direction, another word’s direction and the span of all word directions alike, while a random direction left the text intact. These directions carry large natural variance in this model, so clamping them is not a clean test, and we do not analyze these outputs.

The mean signature is a minor carrier. Subtracting the own-word signature lowers leakage by 5.3 points against no intervention (95% CI [–16.5, +5.8], not significant) and by 10.0 points against subtracting another word’s signature ([–17.6, –3.4], $p = 0.004$, Holm $p = 0.04$; table 4). Doubling the subtraction gives –9.4 points against no intervention ($p = 0.07$). The effect is specific, since the other-word and random controls do not lower leakage, and it comes without a quality cost at the single dose. Leakage stays far above chance in all cases, and the model can still state its secret under every intervention (15 of 15). Adding the signature to a run with no secret produces little leakage (53.1 to 54.4% per trial; 62.2% order-cancelled at 3×). The intervention removes the average effect at each layer; it does not stop later layers from reading the secret again from the prompt tokens.

Removing the secret removes the leak. After a 30-token opening written with the secret, a secret-free continuation is at chance (52.2%; table 5).

The leak does not need the opening. When the first 30 or 100 tokens are the model’s default no-secret opening, a lighthouse keeper in every case, and the secret is present only afterwards, leakage is as high as with the secret throughout (80.6% and 77.5%). The secret keeps entering through the content choices made along the way: a “cactus” story that starts with the lighthouse keeper has him build “a small garden on the leeward side of the tower”.

By 100 tokens some of the theme is in the text. Leakage after removing the secret at 100 tokens is 68.1%. In 20% of those stories the continuation, now free of the instruction, names the secret word outright (for “violin”: “her husband, a renowned violinist”). This fits the readout result that the instruction suppresses the word at the output.

5 Discussion

Does a plan stop the leak? The answer depends on who wrote the plan. If the model wrote it while holding the secret, the plan is where the secret went, and the prose inherits it; this holds for a word and for a whole-story outline that already contains the twist. If the content was decided without the secret, the primary guesser and the text probe find no trace of the secret word in the prose, even though the model still holds the secret and represents it at every story position. For plot twists the same holds in direction, with weaker evidence.

Where does the leak happen? Not in the wording of prose whose content is fixed, and not in a single early commitment. It happens wherever the model chooses content with the secret available: at outline time, and continuously through the body of a free story. The premise and swap conditions together make this point. A two-sentence premise that the model is told to write from stops the leak, while a fixed opening that the model merely continues does not.

Representation versus use. The secret is linearly present at story positions in every condition, at full decodability in late layers. Its mean signature is small, tracks leakage across conditions, and has a modest, specific causal effect, while taking the secret out of the context removes the leak completely. The most economical reading is that the leak is driven by repeated reads of the secret from the prompt as content decisions come up, and that the mean residual signature is a partial summary of those reads. We did not test this reading directly. For monitoring, the consequence is that a probe which finds a secret in the activations says little about whether the output leaks it.

Practical implication. To keep known information out of generated prose, decide the content before the model sees the secret, or in a separate context, and have the model write from that plan. Asking the model to plan for itself does not help.

Limitations. *Guesser.* The primary guesser is the writer model itself, not a frontier model. Its sensitivity is limited, clearly so for twists, so a chance-level result means “not detectable by Gemma 3 12B or by a linear probe on its activations”. The cross-checks are small (49 to 238 trials), their intervals ignore clustering by word, and for the blind outline they point in opposite directions. *Scope.* We study one writer model and 15 words; word-level intervals are wide, and our conclusions rest mainly on paired differences across the same words. The decoy failure and the task-boundary effect may be specific to Gemma 3 12B. *Post hoc additions.* The order-cancelled metric, the 500-token craft filler, the neutral-request filler, the no-hide twist conditions and the swap experiment were added after seeing earlier results. The Holm families include them; the swap experiment is the least pre-specified. *Length.* Outlines were cut at 500 tokens, stories written from outlines are longer (680 words against 524), and 47% of blind-outline stories hit the 900-token cap. Longer stories give a guesser more to work with, so this works against the reduction we report. *Twists.* We wrote the twist stimuli ourselves without human validation, and generated only openings. *Interventions.* Projection ablation did not work without destroying fluency, so the causal evidence comes from mean-shift patching (small effect) and context removal (full effect). Neither isolates a specific component; we did not test attention from story positions to the secret tokens. Quality of intervened stories was rated by the writer model itself.

6 Conclusion

A model holding a secret leaks it into the content decisions it makes, whenever it makes them. Handing Gemma 3 12B a plan helps only when the plan was made without the secret: leakage of a secret word into the prose then drops from 84.1% to about chance, while the secret remains fully decodable inside the model. A plan the model writes for itself absorbs the leak and passes it on. Context that fixes no content changes little, and the leak needs the secret to stay in context while the body of the story is written. For plot twists the direction is the same and the evidence weaker.

Several follow-ups would strengthen these findings. The saved stories should be re-judged with frontier guessers and with humans, especially in the blind-outline and twist conditions. Masking attention from story positions to the secret tokens, layer by layer, would locate the reads that the swap experiment says are necessary. The twist experiment should be run on whole stories with a larger writer. Finally, our results suggest a two-context recipe that we did not test end to end: plan in a context without the secret, then write in one with it.

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A Contrasts for Experiment 1

Contrast	Difference [95% CI]	p	Holm p
BLINDOUTLINE – NOPLAN	−34.4 [−44.9, −24.1]	0.0002	0.003
SELFOUTLINE – NOPLAN	−2.2 [−9.1, +6.8]	0.50	1
BLINDPREMISE – NOPLAN	−29.4 [−42.3, −19.4]	0.0002	0.003
CRAFTFILLER (300) – NOPLAN	−4.6 [−9.6, +0.2]	0.06	0.37
CRAFTFILLER (500) – NOPLAN	−9.0 [−17.3, −1.7]	0.016	0.11
UNRELATEDFILLER (neutral) – NOPLAN	−5.8 [−14.0, +2.6]	0.15	0.59
UNRELATEDFILLER (task switch) – NOPLAN	−28.3 [−40.9, −17.9]	0.0002	0.003
DECOY – NOPLAN	−1.0 [−10.4, +9.8]	0.85	1
BLINDOUTLINE – CRAFTFILLER (500, token-matched)	−25.4 [−36.6, −12.3]	0.0002	0.003
SELFOUTLINE – BLINDOUTLINE	+32.2 [+22.3, +40.8]	0.0002	0.003
UNRELATEDFILLER: neutral – task switch	+22.5 [+14.2, +30.7]	0.0002	0.003

Table 6: Contrasts in per-trial discrimination accuracy (points), paired bootstrap over the 15 words, Holm-corrected within the family. Bootstrap p -values have a floor of 0.0002.

B Cross-checks with other guessers

Check	Result
Nemotron 3 Ultra, word secrets, NOPLAN	89.4% ($n = 132$, Wilson 83.0 to 93.6); Gemma 3 12B on the same trials: 81.8%
Nemotron 3 Ultra, word secrets, BLINDOUTLINE	63.3% ($n = 49$, Wilson 49.3 to 75.3, $p = 0.09$ against 50%); Gemma 3 12B on the same trials: 49.0%
Nemotron 3 Ultra, twists, no plan / twist-blind outline	74.8% ($n = 238$) / 62.8% ($n = 94$)
Gemma 3 27B, word secrets, 192 trials each	NOPLAN 85.9% (Wilson 80.3 to 90.2); BLINDOUTLINE 41.1% (34.4 to 48.2); Gemma 3 12B on the same trials: 83.9% and 47.9%
Gemma 3 27B, twists, 192 trials each, per-trial (order-cancelled)	hide instruction: no plan 64.6% (66.7), twist-blind outline 54.2% (56.2); no hide instruction: 70.3% (72.9) and 60.4% (61.5). It answers “1” in 60 to 67% of trials
Position bias of the Gemma 3 12B guesser	“1” in 42 to 55% of word-secret trials; 89 to 96% of twist trials

Table 7: Cross-checks. Intervals for the subsample guessers are Wilson intervals that ignore clustering by word. The Gemma 3 27B below-chance score on BLINDOUTLINE has binomial $p = 0.02$ without clustering; the two-proportion tests for the twist differences have p of about 0.03 to 0.04 without clustering.

C Per-word accuracy

D Implementation details and deviations from the plan

We ran all experiments on one NVIDIA RTX A6000 (48 GB) with HuggingFace transformers, in about 8.5 GPU hours. Generation used batches of 32 (22 for two-turn prompts) with a base seed of 42. Table 8 lists where the study departed from its plan.

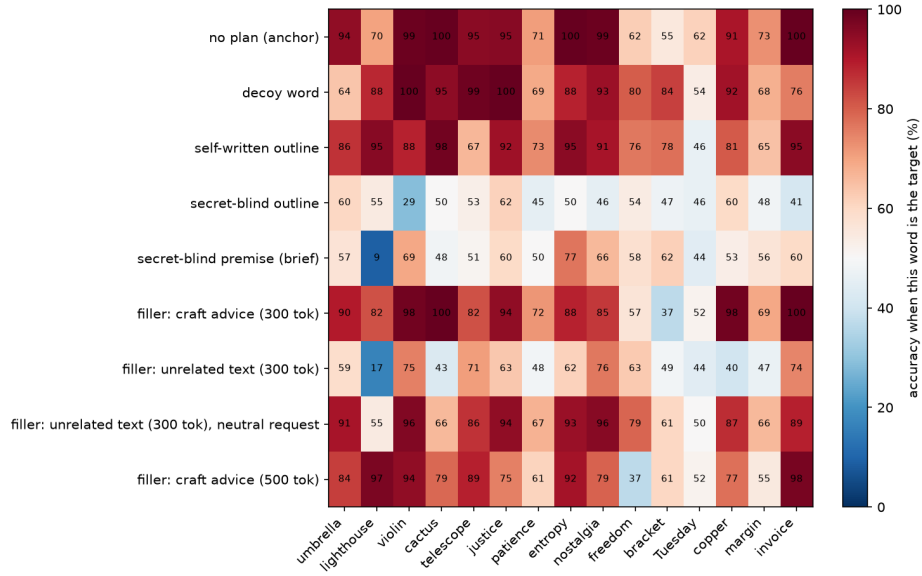


Figure 5: Per-word discrimination accuracy in Experiment 1.

Planned	Done	Reason
Two frontier API guessers	Local Gemma 3 12B (primary); free-tier Nemotron 3 Ultra and local Gemma 3 27B on subsamples	The API returned HTTP 402 (no credits) for all paid models
Per-trial accuracy as the only primary metric	Order-cancelled accuracy reported alongside	89 to 96% position bias in twist trials
Projection ablation of the secret direction	Mean-shift patching and context swap	Projection ablation and clamping produced degenerate text
8 stories per word for interventions; 4 openings per twist	6 and 3	GPU time
Length-matched fillers	First fillers were 300 tokens against 500-token outlines; a 500-token craft filler was added	Found when token counts were checked
LLM judge rating for quality	Rating and likelihood from the writer model	No API credits

Table 8: Deviations from the pre-specified plan.